



Operational Risk Matrix

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Document Version Register

Version Number	Version Date	Approved By	Section Number and Title	Details of Version
V 1.0	10/14/2015	Kori Patrick	LP Risk Matrix (all sections)	Created document to provide guidance on the use of the LP Risk Matrix
V 2.0	05/30/2019	Matthew Thompson	Risk Matrix (all sections)	Document updated to provide guidance on the use of the S&R Standardized Operational Risk Matrix in LP risk assessments
V 2.1	09/03/2020	Matthew Thompson	'Roles and Responsibilities' updated 'References' updated	Updated to reflect implementation of LP Risk Roles
V 2.2	04/15/2023	Matthew Thompson	All sections	Alignment to Framework Risk Management Standard

Purpose

Standardized Operational Risk Matrix shall be used to determine risk levels for a given asset, project, or activity within Liquids Pipelines (LP). The Standardized Operational Risk Matrix expresses the risk in our business through the qualitative ranking of risk using the likelihood and potential consequences of events. The Standardized Operational Risk Matrix provides guidance regarding the required level of risk authorities for escalation and risk acceptance based on the assessed level of risk.

Scope

Executive Leadership through the Safety and Reliability Governance Team (SRGT) authorize the Standardized Operational Risk Matrix. It is used in the assessment of hazards and risks that may impact Workforce Safety, Public Safety, Environmental Incident, Operational Reliability, Financials, and Enbridge's Reputation. It applies to all business units and their functions including departments, operating regions, and projects.

The risk matrix is ubiquitously integrated into LP business applications, tools, and processes. It can also be used for ad-hoc risk assessments and operational decision making as required. [Risk Assessment Tools and Methodologies](#) provides a summary of the primary instances of use of the Standardized Operational Risk Matrix in LP.

Determination of a high or very high risk using the Standardized Operational Risk Matrix often warrants a more involved or advanced method of risk assessment. If unclear in these cases, engage with LP Risk Management to identify a fit for purpose solution using either in house resources or specialized external services.

Roles & Responsibilities

See [LP Risk Roles](#) for normative role expectations in risk management across the LP organization. The following section outlines the responsibilities between corporate and business functions in stewardship of the operational risk matrix.

Safety and Reliability	- Responsible for stewardship of the Operational Risk Matrix and its alignment across all business units as expressed in the Framework Risk Management Standard - Administers and maintains the Standardized Operational Risk Matrix current with business conditions - Maintains governance and approval of alternate risk matrices used in corporate and central functions outside the scope of Liquid Pipelines operational risk management
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LP Risk Management	- Provides LP input towards the risk characterization defined in the Framework Risk Management Standard. Assures integration of its content into LP and alignment to the Hazard and Risk Management Process - Defines Risk Criteria and Risk Authorities the LP business unit
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Standards and Requirements

The Risk Management process as documented within the Integrated Management System (IMS) refers to the ISO 31000 standard as a baseline and aligns Enbridge with Risk Management regulations including, but not limited to the Canada Energy Regulator (CER) Onshore Pipeline Regulations, the U.S. Department of Transportation Pipeline and Hazardous Materials Safety Administration (PHMSA) Title 49 CFR Part 192 regulations; as well as consideration of industry standards including, but not limited to the Canadian Standards Association (CSA) Z662 standards, and American Petroleum Institute (API) 1173 Pipeline Safety Management Systems Recommended Practice. The use of a consequence/probability matrix as a risk assessment tool is described in CSA-IEC/ISO 31010-20 – Risk Management – Risk Assessment Techniques.

General

Enbridge Liquids Pipelines uses the Standardized Operational Risk Matrix (Figure 1) to assess the risk levels of hazards or other risk sources based on standardized levels of potential consequence and likelihood of occurrence of a given scenario.

Standardized Operational Matrix Use

To use the Standardized Operational Risk Matrix requires an understanding of the specific hazard, risk, and event that is to be assessed. General guidance regarding consequence analysis, likelihood estimation, and the consideration of controls are contained within the **Risk Analysis** Process, while best practice and normative expectations specific to a given application of the risk matrix is found in process and procedure documents. See **Risk Assessment Tools and Methodologies** for a summary of the primary instances of use of the Standardized Operational Risk Matrix in LP.

To establish Risk Level, each matched pair of consequence and likelihood ratings is plotted on the Risk Matrix. The location of the plotted risk on the Risk Matrix identifies the risk level. For guidance on risk reduction requirements, escalations, and timeframes for action based on level of assessed risk see **Risk Evaluation**.

Note central and corporate functions, such as TIS and Projects, have exclusions to use a separate (5x5) risk matrix for their internal use for their business function. The expectation is any service interaction with the LP Business Unit requires application of the 7x7 Standardized Operational Risk Matrix.

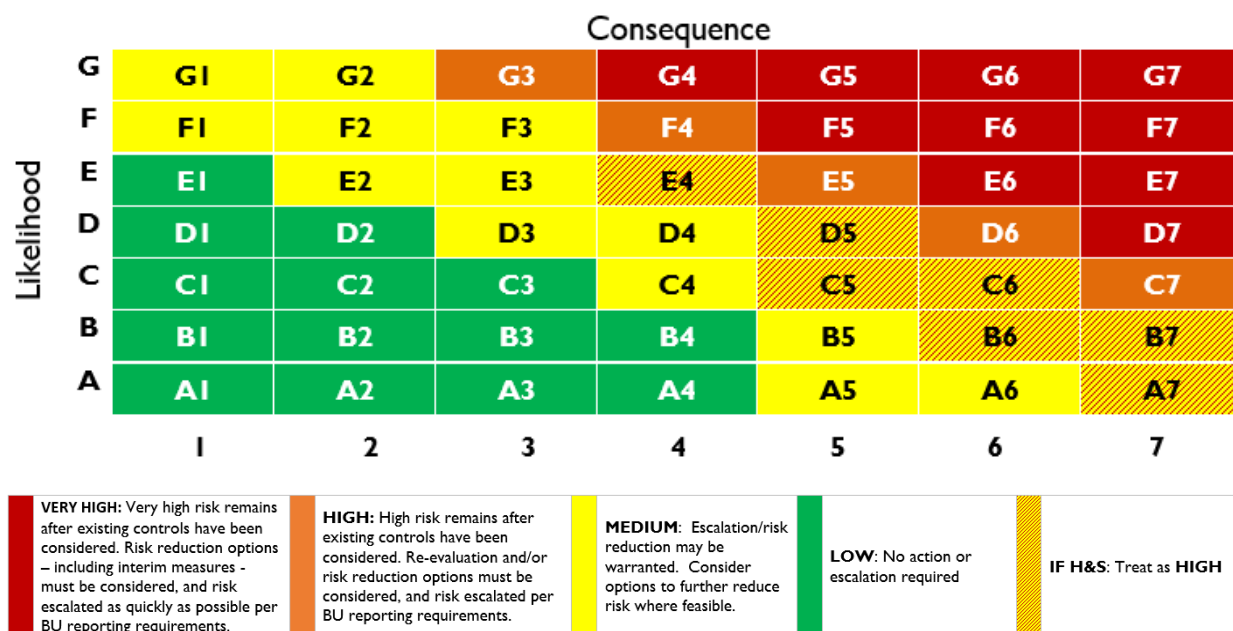


Figure 1 – Standardized Operational Risk Matrix

To ensure consistency of assessment, the Standardized Operational Risk Matrix provides descriptions of both consequence and likelihood levels for all risk receptors of principal concern to Enbridge. Interpretation may be required to address certain scenarios. Figure 2 provides details regarding Enbridge risk receptors. Consequence and Likelihood levels are described in detail in Table 1 and Table 2, respectively.

Receptor	Description
Financial	Financial impacts may include clean up and remediation costs associated with operational events, revenue losses, fines, etc. When assessing financial impacts currency should be mindful of the location of the risk e.g., USD vs. CAD.
Health and Safety	The risk matrix cannot describe all potential health impacts associated with an operational incident. The terms and definitions below are intended to provide context but should not be considered exhaustive. First aid: Minor abrasions and minor bruises

	Minor Injuries: Short recovery; minor lacerations, minor burns, and minor sprains/strains Moderate Injuries: Moderate recovery (weeks to months); fractures, 2nd/3rd degree burns, and significant strains/sprains Major Injuries: Long-term/life altering; life-altering fractures, significant third-degree burns, disfigurement, and limb-loss Note that cells E4, C5, D5, C6, B6, B7, and A7 shall be treated as “High” risk when assessing safety-related risks. Consequence 5 or higher for Health & Safety requires assessment and validation in the use of this qualitative risk matrix. Contact LPRM for additional information.
Environmental	Consequence descriptions reflect increasing sensitivity of potentially affected areas with a focus on land and water impacts with descriptive relevance for offsite impacts. It is important to consider magnitude of impact as well as the nature of the receptor when assessing environmental risk e.g., in the case of a liquid hydrocarbon release, the potential volume of a release must be considered along with the sensitivity of the receptor to derive a representative risk level. The consequence descriptions may be applied to many mechanisms of environmental damage e.g., crude oil releases, fire damage, non-hydrocarbon liquids releases, etc.; however, consideration should be given to the mechanism of damage and scaling applied e.g., identical volumes of crude oil and wastewater may have very different environmental impacts in the same location. Risk associated with non-land/water-based contaminants (e.g., unplanned air emissions, noise, odor) should also be considered; however, given the relative potential for environmental damage from, for example, air-release events compared to other mechanisms such as liquids releases or fire, other consequence categories may be more appropriate e.g., impacts such as fines, shutdowns, or reputational consequences associated with air emission events; and/or H&S consequences associated with potential toxicity of emissions.

	For onsite impact LP provides direction for release within an Enbridge facility as seen in Table 1. Consequence 5 or higher for Environment requires assessment and validation in the use of this qualitative risk matrix. Contact LPRM for additional information.
Operational	When assessing potential operational impacts, consideration should be given to the magnitude, duration, and type of customer or organizational impact that an operational interruption may create. LP provides additional direction of suggested severity in duration of operational impact events as seen in Table 1.
Reputational	When assessing potential reputational impacts, consideration should be given to the visibility of the potential event i.e., remoteness of affected location; any cultural/social significance associated with the area that the event may occur in (e.g., parkland/public spaces, Indigenous lands, cemeteries etc.); and any existing or anticipated sensitivity associated with the affected asset. i.e., assets currently under media, public, or regulatory scrutiny are more likely to cause significant reputational impacts if they are involved in an incident. These factors may influence the media and regulatory descriptions provided in the matrix. For any non-compliance contact regulatory affairs.

Figure 2 - Enbridge Risk Receptors

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	1	2	3	4	5	6	7
FINANCIAL	Total financial impact ≤\$10,000	Total financial impact >\$10k and ≤\$100k	Total financial impact >\$100k and ≤\$1M	Total financial impact > \$1M and ≤ \$10M	Total financial impact >\$10M and ≤\$100M	Total financial impact >\$100M and ≤\$1B	Total financial impact of >\$1B
HEALTH & SAFETY	0 to 10 first aids	1 to 10 minor injuries	1 to 10 moderate injuries	1 to 10 major injuries	1 to 10 fatalities (Note 1)	10 to 100 fatalities (Note 1)	> 100 fatalities (Note 1)
ENVIRONMENT	Barren land or land used for industrial purposes where a spill would have no potential for water resource impact; not exceeding 1 m3	Grassland, grazing areas, or forested areas where a spill would have minimal potential for water resource; impact not exceeding 1 m3 Impacts to surface gravel or soil within an Enbridge facility; able to be remediated by trained personnel quickly and effectively (Note 2)	Agricultural areas where a spill would have minimal potential for water resource, impact, not exceeding 1m3	Cropland, grassland, or forested areas where a spill would have significant potential for water resource impact; not exceeding 1m3. Or environmental consequence 1,2,3,4 from 1 m3 to 10 m3	Populated areas containing ecologically or socioeconomically important spill sensitive resources; or similarly sensitive water bodies including water that serves as a major drinking and/or food source. Or any release 10 m3 to 100 m3 (Note 1)	Areas with very important ecologically or socioeconomically spill sensitive resources; or similarly sensitive water bodies that include habitat for threatened species. Or any release 100 m3 to 1000 m3 (Note 1)	Areas with extremely important ecologically or socioeconomically spill sensitive resources; or similarly sensitive water bodies including those that serve as endangered species habitat. Or any release > 1000 m3 (Note 1)

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OPERATIONAL	No significant capacity disruption	Minor reduction in capacity/ability to deliver	Significant reduction in capacity/ability to deliver From a day to as much as one week (Note 2)	Loss of a major asset/facility for a short period of time From a week to one month (Note 2)	Inability to operate or loss of a major asset/facility for an extended period From 1 to 3 months (Note 2)	Loss of multiple major asset/facilities for an extended period From 3 to 6 months (Note 2)	Loss of multiple major assets/facilities for an indefinite or permanent period Greater than 6 months (Note 2)
	No significant media coverage No unplanned regulatory engagement	Isolated individual concern at municipal/county level. No media attention. Regulator notification and/or informal and unplanned meetings or information requests from regulator; no monetary penalty imposed	Localized concern with short term local media and interest group concerns Non-compliance identified by a regulator in writing without a monetary penalty; may require corrective actions; follow up communication with the regulator regarding the issue expected	State/Provincial concern, public and media attention beyond local area, customer attention on the issue Non-compliance identified by a regulator in writing including a monetary penalty; may require corrective actions; follow up communication with the regulator regarding the issue expected; permit/approval conditions or approval agency change causing moderate operational impacts	National concern and extended media coverage; significant public response causing major impact on current and prospective customers Non-compliance identified by a regulator in writing requiring significant corrective actions; may include a monetary penalty; operating permit/approval suspended causing significant operational impacts	Extended national media coverage; significant public response causing long term impact on customers; inability to expand operations Non-compliance identified by regulator in writing directing Enbridge to stop operating specific assets; may include criminal prosecutions; operating permit/approval canceled causing indefinite suspension of operations	Extended national media coverage; severe public response causing potentially permanent impact on customers; irreparable reputation damage resulting in the inability to continue operations Unable to gain regulatory approval for continued operation; may require decommissioning of major facilities; criminal prosecutions
REPUTATIONAL							

- Consequence 5 or higher for Health & Safety or Environment requires assessment and validation in the use of this qualitative risk matrix. Contact LPRM for additional information.
- LP specific guidance added to Operational Risk Matrix

Table 1 - Consequence Descriptions

	Description	Annual Likelihood	
G	One or more events expected per year at a given facility	≥ 1	≥ 1
F	Event expected several times in a business unit in a one-year period	1/10 to 1	10^{-1} to 1
E	Event expected several times across Enbridge over a one-year period	1/10 to 1/100	10^{-1} to 10^{-2}
D	Isolated prior or expected cases at Enbridge	1/100 to 1/1,000	10^{-2} to 10^{-3}
C	Expected to occur periodically in industry; limited cases at Enbridge	1/1,000 to 1/10,000	10^{-3} to 10^{-4}
B	Known to have happened once in the industry	1/10,000 to 1/100,000	10^{-4} to 10^{-5}
A	No known prior occurrences in industry; unlikely to occur at Enbridge	$< 1/100,000$	$< 10^{-5}$

Table 2 – Likelihood Descriptions

Related Documents

Framework Risk Management Standard	Elink
Hazard and Risk Management Process	GDL
Risk Criteria - Operational Risk.....	GDL
LP Risk Roles	GDL
Risk Evaluation	GDL
Risk Analysis	GDL
Risk Assessment Tools and Methodologies	GDL

References

CAN/CSA-IEC/ISO 31000.....	IHS
CAN/CSA-IEC/ISO 31010.....	IHS

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